Mr Dimelow

Smart and Modern Materials

Topic	What students need to learn:
To apply a knowledge and understanding of **working properties**, **characteristics**, **applications**, **advantages and disadvantages** of the following types of materials in order to discriminate between them and select appropriately.	
1 Materials	1.7 - Smart and modern materials: *a) thermo-ceramics* *b) shape memory alloys (SMA)* *c) reactive glass* *d) liquid crystal displays (LCD)* *e) photo-chromic materials* *f) thermo-chromic materials* *g) quantum tunnelling composites*

What are *'smart materials?'*

Definition:

Smart materials are materials that have **properties which change reversibly**, i.e. can change easily but can then easily change back, depending on **changes in their surroundings**.

Smart materials can respond to ***changes*** in *stress, moisture, electric or magnetic fields, light, temperature, pH, or chemical compounds*.

What are *'smart materials?'*

1. Using the ***key terms*** below, define the term ***'smart material'*** in a few words:

Change, property, significant, reaction, temperature, light

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Advantages of *'smart materials?'*

With benefits for aerospace, medical, textile, construction, and electronics industries, smart materials improve efficiency and save resources by responding to corrosion, pH changes, water content, temperature, mechanical forces, and much more.

What are *'smart materials?'*

2. Describe ***TWO*** general ***advantages*** to the consumer of smart materials-based products

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Disadvantages of *'smart materials?'*

- Can be expensive materials to develop and produce.
- May have lower value mechanical properties compared to products produced using more conventional materials.
- Products may have limitations in terms of their performance.
- The chemicals used to make some smart material based products may be difficult to break down by nature.

What are *'smart materials?'*

3. Describe ***TWO*** general ***disadvantages*** to the consumer of smart materials-based products

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T*hermo-Ceramics*

What are *'thermo-ceramics?'*



They are a ***mixture*** of ***metallic powder*** and ***ceramic powder combined*** using ***sintering***.

What is sintering?

It is when two or more (powdered) solid materials are compacted into a mould and formed by heating up the mould and applying pressure. The amount of heat and pressure applied during the sintering process is slightly less than the material's melting point.

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What is good about *thermo-ceramics structure?*
It offers great hardness and stability, even at high temperatures of 1000°C.

What are *thermo-ceramics used for?*
Turbine blades, automotive parts, high quality (skateboard) bearings and metal cutting tools.







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Advantages of *thermo-ceramics:*

- *It has high heat resistance*
- *High strength*
- *It doesn't expand or deform – so it is a stable material*
- *It is hard – and has excellent durability/wear resistance*
- *It is lightweight*

Disadvantages of *thermo-ceramics:*

- *It is brittle – so can shatter when it is dropped*
- *It is an expensive material to develop and produce*

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Video: Thermo-ceramics Bearings



Shape Memory Alloys (SMA)

What are *'shape memory alloys (SMA)?'*

Shape Memory Alloys (SMA) are metals that have been designed to work in a particular way in response to stresses and strains placed on them.

The most common is called 'nitinol': an alloy of nickel and titanium.

Other SMA can contain a range of other materials such as iron, nickel, cobalt and titanium alloys in varying percentages.

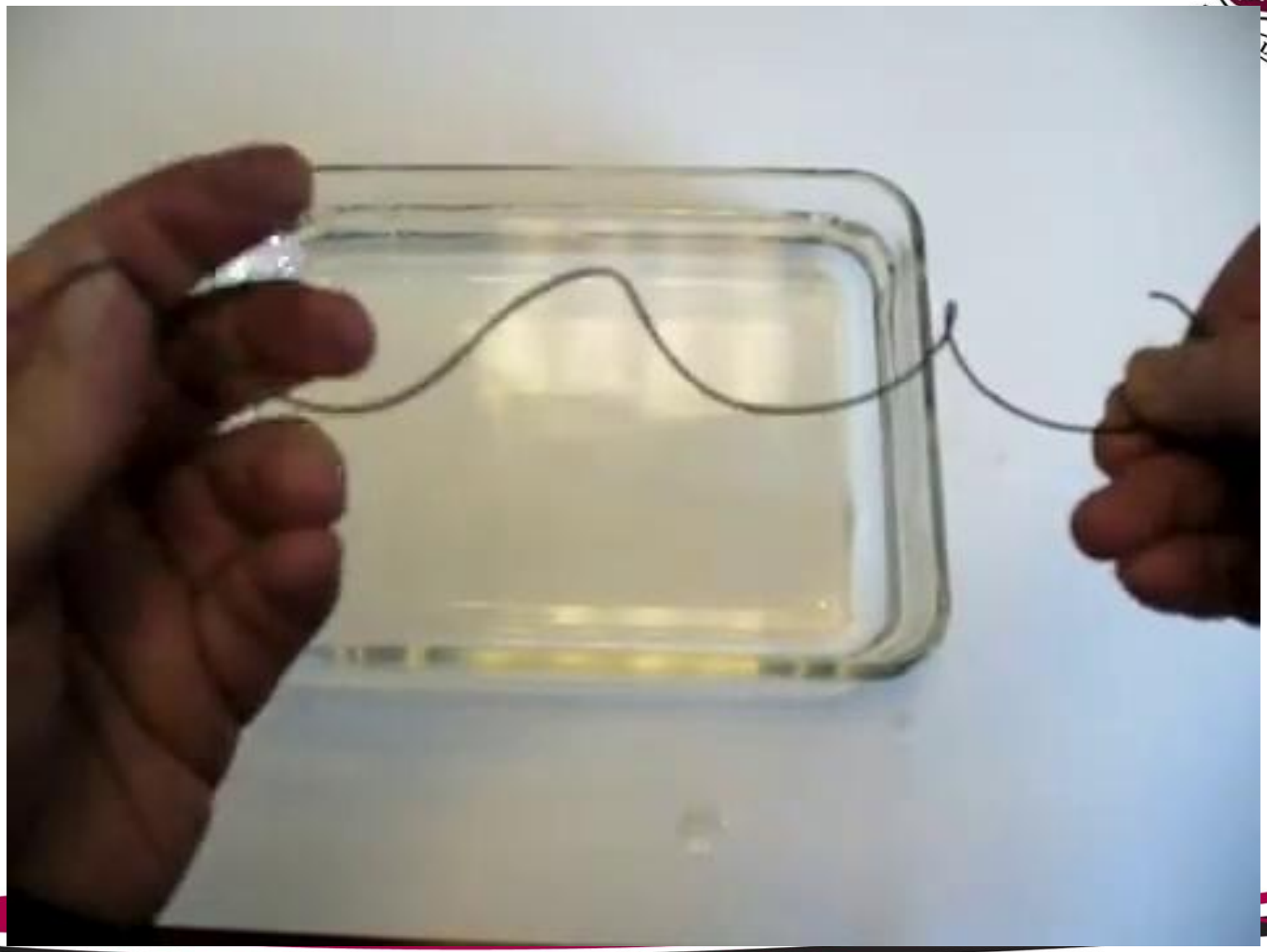
What are *'shape memory alloys (SMA)?'*

When made into components these alloys can be 'plastically' deformed at a certain temperature.

They will retain their shape while held at this temperature. When the temperature is removed the component reverts to its original shape.

It was discovered that by processing alloys in certain ways some alloys could be 'trained' to show memory where the components demonstrated total reversibility on heating and cooling.

Video: *Shape Memory Alloys (SMA)*



Applications - *Muscle Wire*

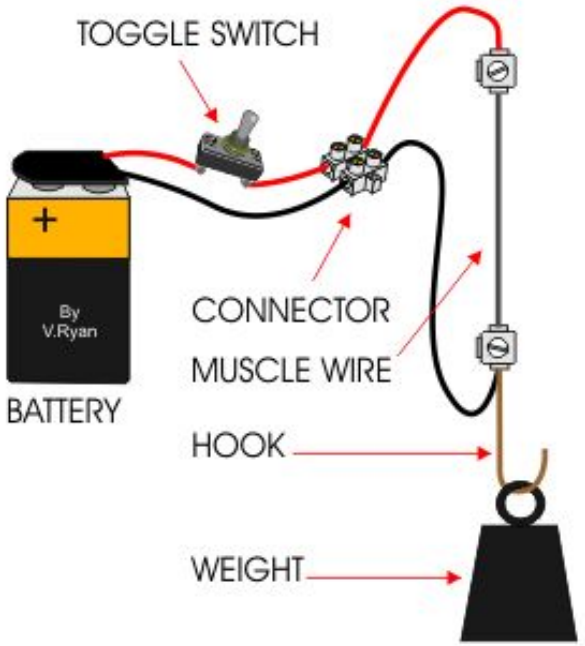
Muscle wire is also a nickel and titanium alloy. At room temperature it can be stretched by a small force. However, when a small current is passed through the wire it returns to a much harder form and to its original length with a reasonable force.

When in use a muscle wire can be stretched up to 8 percent of its length and still recover. However, this can only be done a few times until it breaks or stops returning to its original length. Its life cycle can be extended dramatically if it is stretched to between 3 to 5 percent of its overall length. Within this range it will go through the stretching and return cycle millions of times.

http://www.technologystudent.com/equip1/sma2.htm

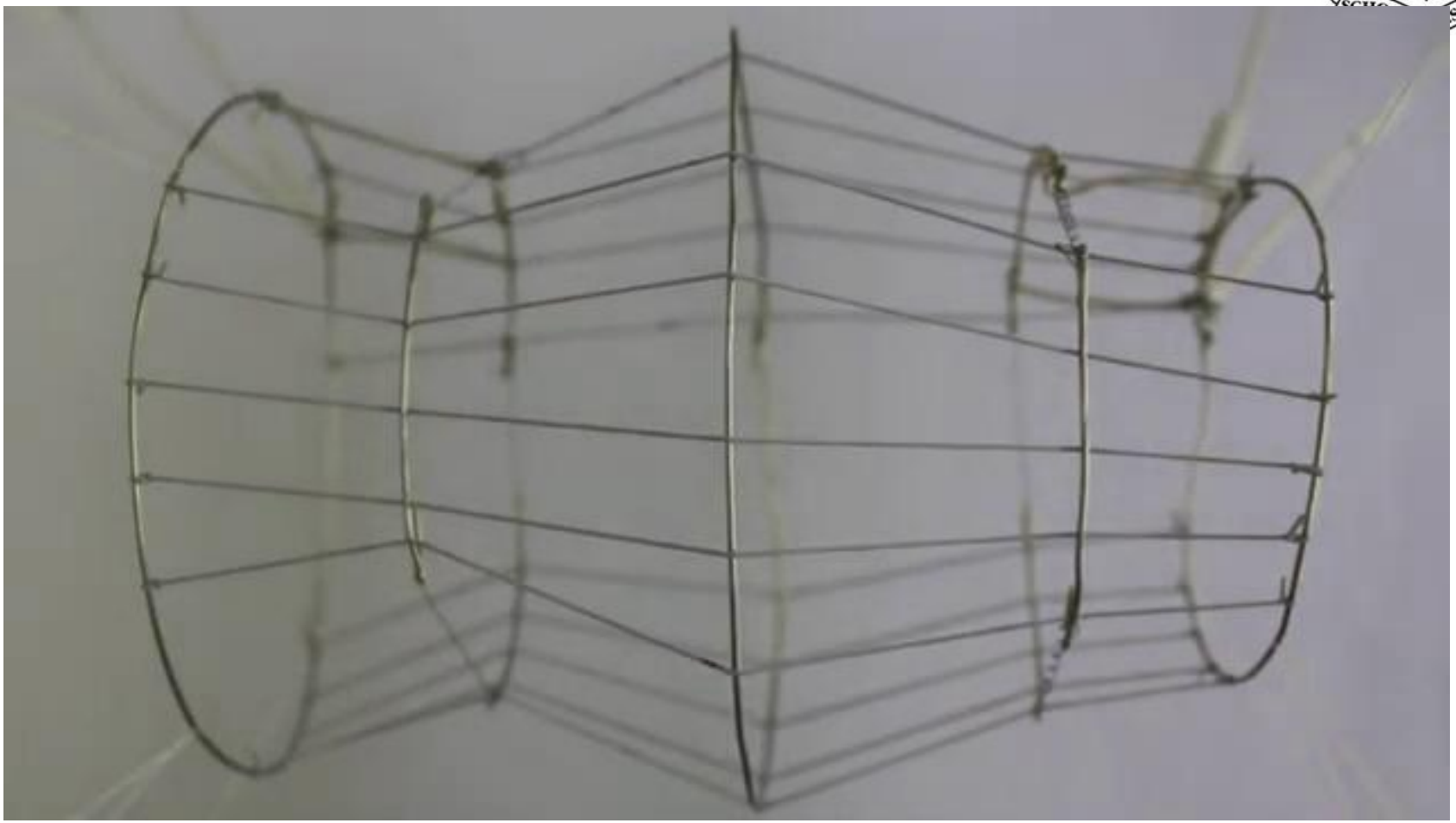
Applications - *Muscle Wire*

The diagram opposite shows a battery and switch connected to muscle wire. A small weight stretches the muscle wire approximately 3 to 5 percent of its length. However, when a current is applied to the wire, it shortens lifting the weight. This cycle of turning on and off the current has the effect of lifting and then lowering the weight.



http://www.technologystudent.com/equip1/sma2.htm

Video: *Muscle Wire*

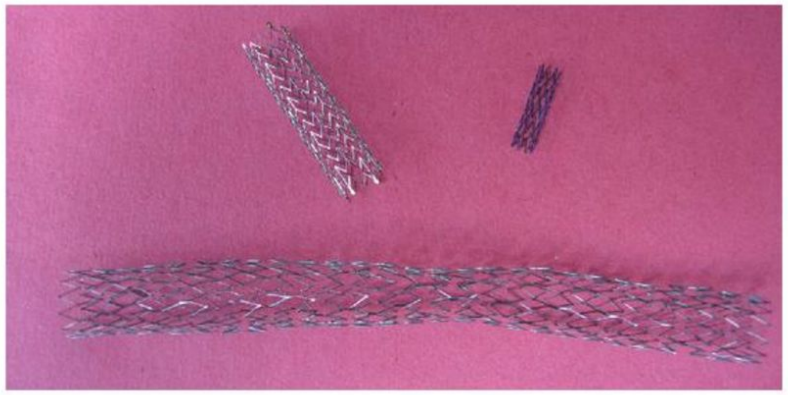


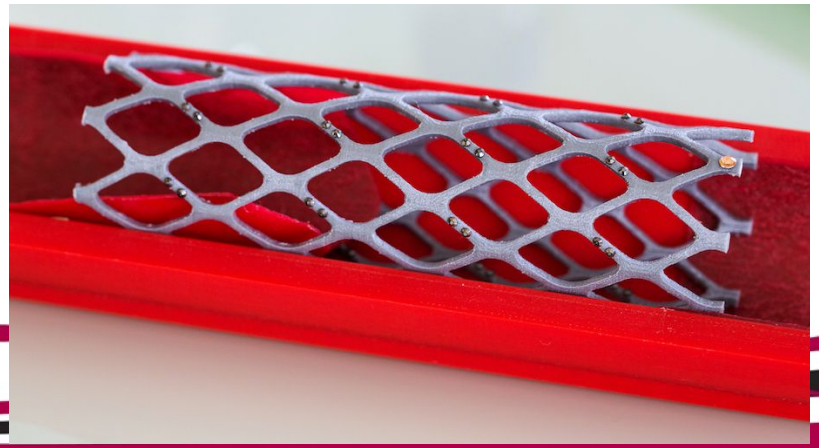
Applications - *Bio-Engineering*

Using heat treatment to give SMA materials a ‘memory’ makes it useful for applications in ***bio-engineering***.

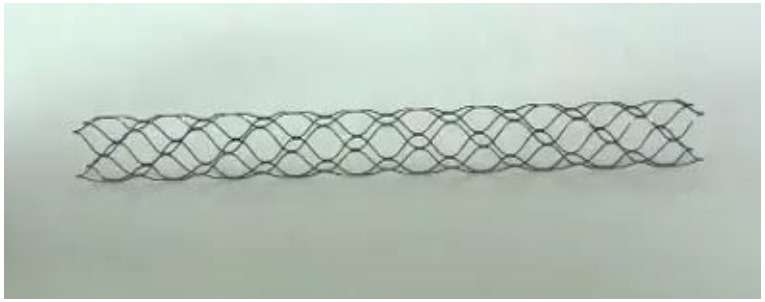
For example, a ‘Stent tube’ (a nitinol tube in the form of a mesh) can be collapsed when ***chilled***.

When inserted into a blocked or collapsed artery it will resume its ***original shape*** as it reaches body temperature. Therefore allowing blood to flow through the artery again.





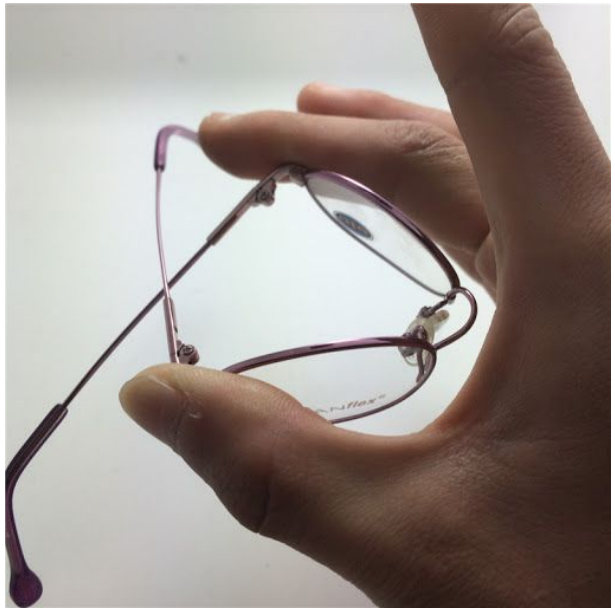
Video: *Bio-Engineering – Stents Tubes*





Applications - *'Memoflex'* Glasses

'Memoflex' glasses are a further example of the use of SMA.



These are nickel/titanium alloys that have the ability to regain their original shape at room temperature after being deformed.

These are termed 'super-elastic' alloys. Deformation of these materials can be complex, e.g. forming a wire into a knot, while the material is capable of complete recovery.

Advantages of shape memory alloys (SMA)

- Shape memory alloys are a unique class of alloys that have ability to 'remember' their shape and are able to return to that shape even after being bent.
- Good mechanical properties – strong, corrosion resistant.
- At a low temperature, a SMA can be seemingly plastically deformed, but this 'plastic' strain can be recovered by increasing the temperature.
- Diverse range of applications. Including Bio-engineering compatibility – safe to use.

Disadvantages of shape memory alloys (SMA)

- SMA are still relatively expensive to manufacture and machine compared to other metallic materials like steel and aluminium alloys.

- Most SMA have poor ‘fatigue’ properties. This means under the same loading conditions (twisting, bending, compression) a steel component may survive more cycles than a SMA component.

Reactive Glass

What is *'reactive glass?'*

Reactive glass turns *opaque / clears* when an ***electrical current*** is applied providing immediate change between privacy / protection and clear glass ***unlike*** *photochromic glass which slowly tints in reaction to UV light* – and is not controllable.

The application of current / opacity can be automatically controlled as ***switching*** can be via ***control circuits*** which can ***react*** to external stimuli via ***light sensors***.

What is *'reactive glass?'*

Reactive glass can be used to provide ***user safety*** by *reducing light to a safe level that will not damage eyesight / distract the user* in applications such as *auto-dimming rear view mirrors, welding visors etc.*



https://www.autoworldjournal.com/smart-glass-windows-revolutionizing-the-future-of-driving/



https://www.weldingandwelder.com/product/speedglas-9100xx-auto-darkening-welding-helmet/

What is *'reactive glass?'*

There is no requirement for ***curtains/blinds*** providing *initial/replacement* ***cost savings*** - allowing a *choice of aesthetic approach and ongoing savings on maintenance/cleaning*.



https://www.explainthatstuff.com/electrochromic-windows.html

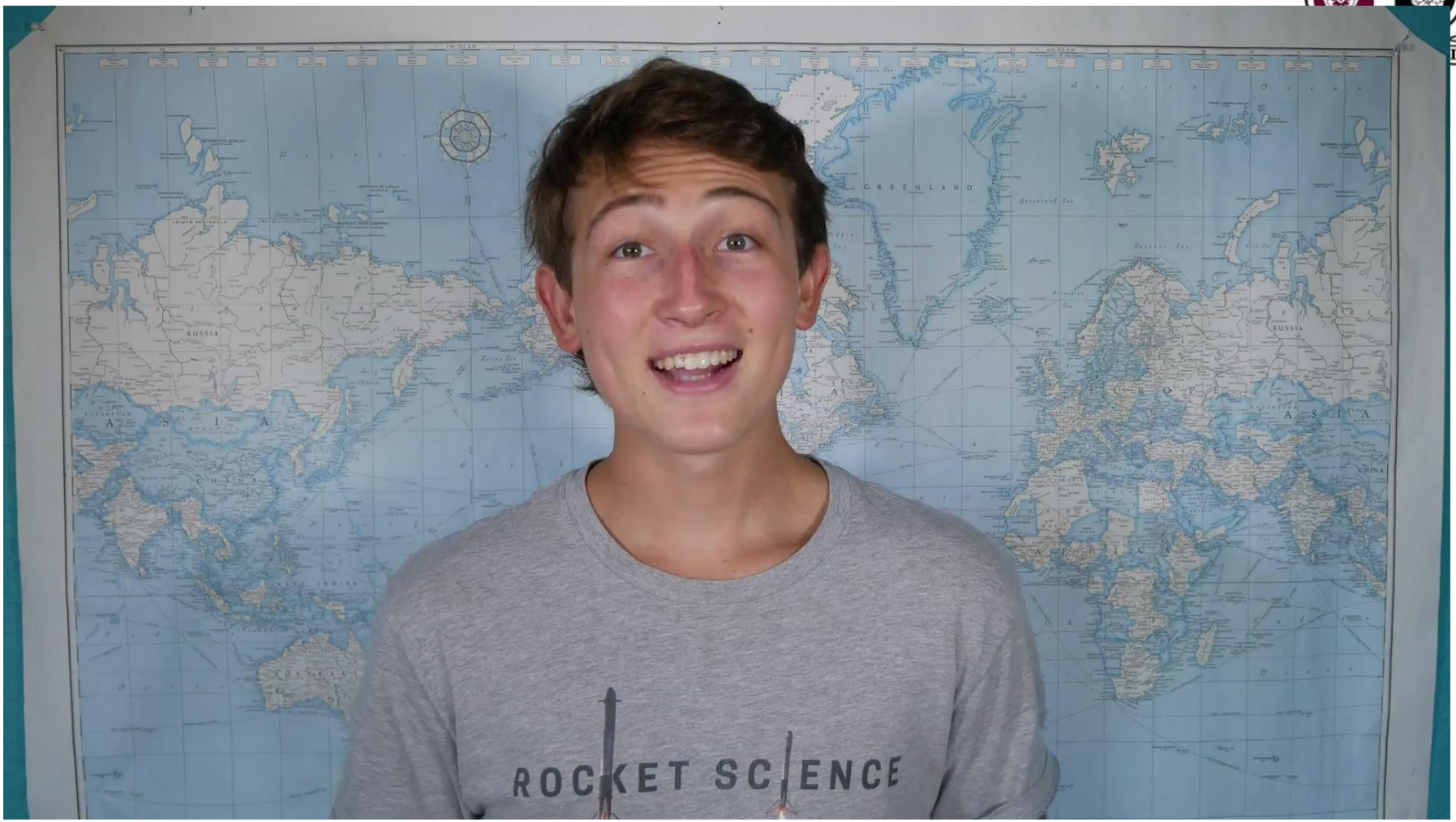
https://www.iqglassuk.com/products/electrochromic-glass/s62640/

What is *'reactive glass?'*

Reactive Glass allows the ***control*** of *solar heat gain reducing air conditioning costs* and *providing a more comfortable user environment / reducing environmental impacts.*

Changes instantly from clear to opaque and opaque to clear making it ideal for situations where privacy is needed therefore allowing more efficient working / use of the building.

Video: The Science Behind Smart Glass



https://www.youtube.com/watch?v=7yFXKHV_MkU

Liquid Crystal Displays (LCD)

What is a *'Liquid Crystal Display (LCD)?'*

LCD (Liquid Crystal Display) is a type of flat panel display which uses liquid crystals in its primary form of operation.

LCDs have a large and varying set of use cases for consumers and businesses, as they can be commonly found in smartphones, televisions, computer monitors and instrument panels.

https://whatis.techtarget.com/definition/LCD-liquid-crystal-display

What is a *'Liquid Crystal Display (LCD)?'*

LCDs were a big leap in terms of the technology they replaced, which include light-emitting diode (LED) and gas-plasma displays. LCDs allowed displays to be much thinner than cathode ray tube (CRT) technology. LCDs consume much less power than LED and gas-display displays because they work on the principle of blocking light rather than emitting it. Where an LED emits light, the liquid crystals in an LCD produces an image using a backlight.

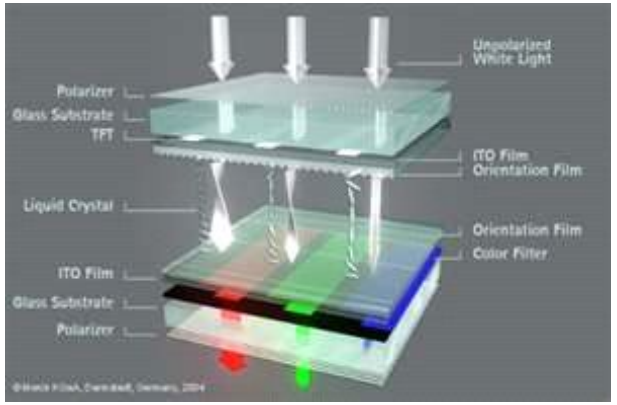
https://whatis.techtarget.com/definition/LCD-liquid-crystal-display

What is a *'Liquid Crystal Display (LCD)?'*

As LCDs have replaced older display technologies, LCDs have begun being replaced by new display technologies such as OLEDs.





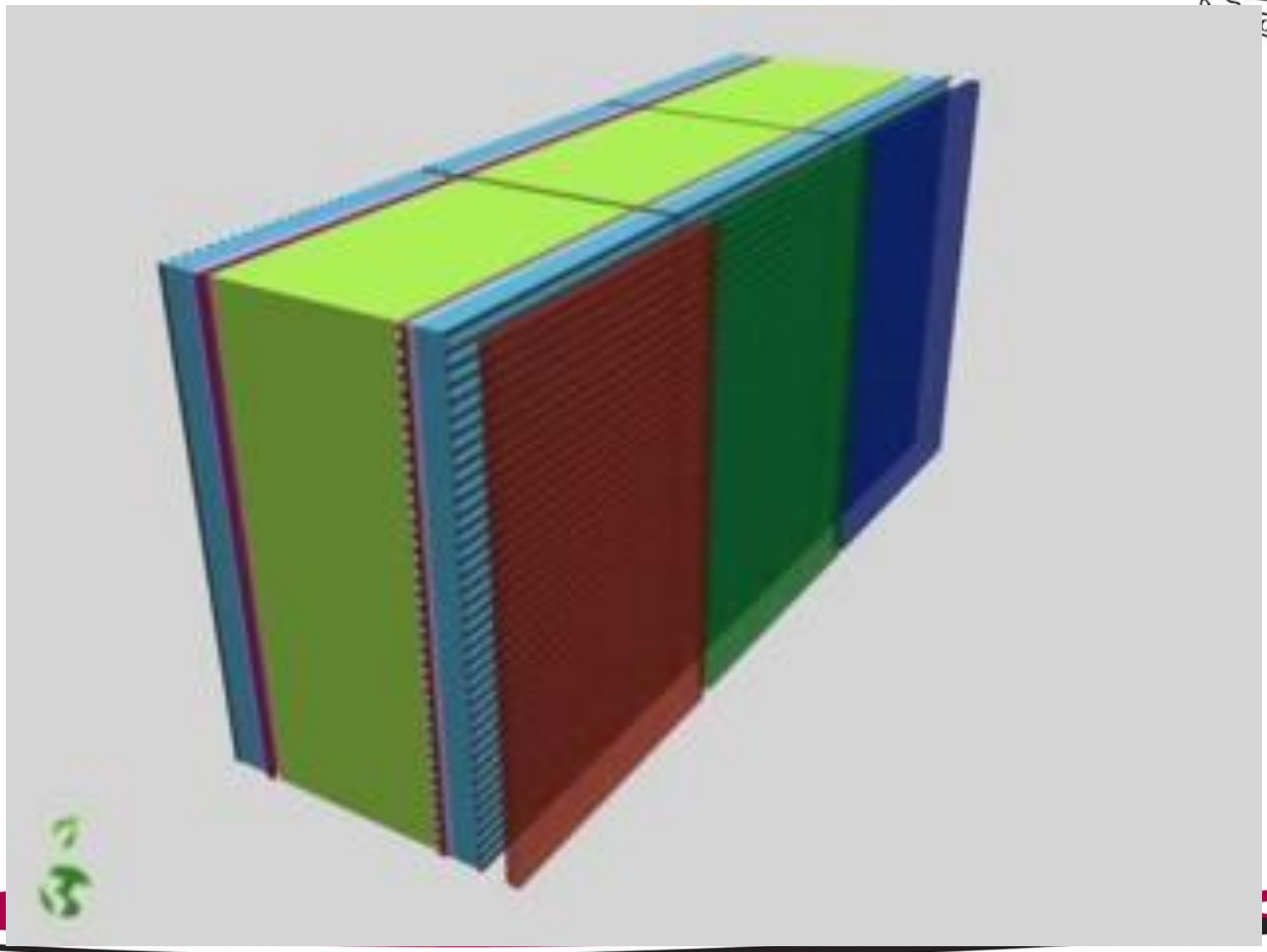


https://whatis.techtarget.com/definition/LCD-liquid-crystal-display

How does a *'Liquid Crystal Display (LCD)' work?*

Liquid crystal display technology works by blocking light. Specifically, an LCD is made of two pieces of polarized glass (also called substrate) that contain a liquid crystal material between them. A backlight creates light that passes through the first substrate. At the same time, electrical currents cause the liquid crystal molecules to align to allow varying levels of light to pass through to the second substrate and create the colours and images that you see.

Video: *Liquid Crystal Display (LCD)*



Applications - *'Liquid Crystal Display (LCD)'*

LCDs are used in a wide range of applications, including LCD televisions, computer monitors, mobile phones, instrument panels, aircraft cockpit displays, and indoor and outdoor signage.

Advantages - *'Liquid Crystal Display (LCD)'*

- The LCD can be made in large sizes of over 60 inch or 1500 mm diagonal.
- It has no geometric distortion.
- It is not affected by the magnetic fields.
- Due to low power consumption, small heat emitted during operation.
- It is much thinner and lighter than a CRT (cathode ray tube) display/monitor.

Disadvantages - *'Liquid Crystal Display (LCD)'*

- In high temperature environments there is loss of contrast.
- It is relatively bright but not suitable for very brightly environments.
- It has individual liquid crystals which cannot completely block out the backlight.
- From the viewing angle, the colour and contrast not consistent.
- LCD displays are expensive.

Photo-Chromic Materials

What are *'photo-chromic materials?'*

Photochromic ink darkens, as the light level increases. Some photochromic inks change colour. In fact, it is UV light that causes the darkening of the ink, which means the ink works best in natural light. This special ink has two main applications; sunglasses and spectacles. However, it is also used in novelty items such as embroidery thread and toys, where colour change takes place according to light level. Nail lacquer / varnish is also available with photochromic technology.

http://www.technologystudent.com/joints/photchrom1.html

What are *'photo-chromic materials?'*



The can on the left shows what the design looks like under indoor lighting; the can on the right shows how the photochromic inks respond to sunlight.

https://www.bakeryandsnacks.com/Article/2015/01/07/photochromic-inks-sunlight-metal-deco-can-manufacturing

Video: *Photo-Chromic Labels*



Phosphorescent Pigments

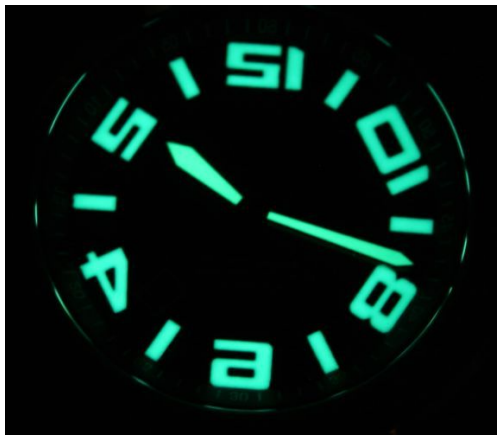
Phosphorescent Pigments were originally developed as a replacement for the more dangerous radioactive materials and older phosphorescents used in the manufacture of glow-in-the-dark watch and clock hands.

A new ceramic material has been developed that will absorb large amounts of light energy over a period of time.

The pigments are available as a powder, and can be mixed with acrylic paints and inks to produce signs that can be seen in complete darkness.

Applications - *Phosphorescent Pigments*

Examples of their use include watch faces that can be read 24 hours a day.



Safety signs, which must be visible in murky / dark environments (e.g. a sign pointing towards a fire escape / exit) – use Strontium Aluminate.





Advantages of *Phosphorescent Pigments*

- Store light for several minutes or even hours and re-emit a significant amount of light over a long period of time.
- Can be recharged when exposed to light or other electromagnetic wave.
- Relatively inexpensive and can be used in the form of a powder, gel, liquid, or gas.

Disadvantages of *Phosphorescent Pigments*

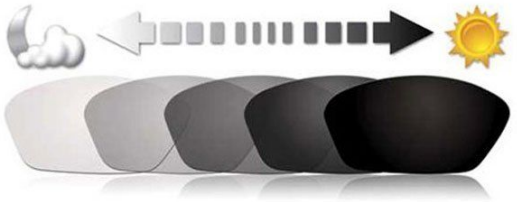
- Cannot be completely sure how bright a phosphorescent pigment is or how long it will emit light.

What are *'reactive glass?'*

A 'reactive glass' is used to produce photochromic lens (or transition lens, as a generic trademark from the brand name Transitions).

These are optical lens that darken on exposure to specific types of light of sufficient intensity, most commonly ultraviolet (UV) radiation. In the absence of activating light the lenses return to their clear state.





https://en.wikipedia.org/wiki/Photochromic_lens

Video: R*eactive glass*



https://en.wikipedia.org/wiki/Photochromic_lens

How is *'reactive glass' made?*

It is made like normal glass but it also contains 'silver-halide'.

Silver-halide reacts with UV radiation and changes size by a fraction of a nanometre.

This small changes adds up when there are millions of silver-halide particles.

Applications of reactive glass:
Everyday glasses (reading/sun glasses), motor bike helmet visors, windows.

https://quizlet.com/96827172/modern-materials-and-products-flash-cards/

Advantages of *'reactive glass'*

- They can provide the correct amount of protection for your eyes at varying light concentrations and it absorbs 100% of UV radiation, so your eyes are not damaged.

- Replaces the need for separate reading and sunglasses

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Disadvantages of *'reactive glass'*

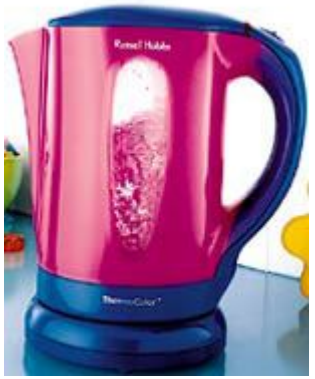
- Expensive to manufacture
- Smart glass is expensive to install
- Time delay of photochromic glasses can cause difficulties when driving.

Thermo-Chromic Materials

What are *'thermo-chromic materials?'*

Thermo-chromic pigments are colour pigments that can change colour in response to heat. The pigments are typically combined with polymers as plastic products are moulded.

The Russel Hobbs 2001 'Thermocolour' kettle, often referred to as the 'pink kettle', changes from a cool blue colour when cold to a vibrant pink as it boils.





What are *'thermo-chromic materials?'*

Baby feeding products, such as bowls, spoons and cups by Tommee Tippee, use thermo-chromatic pigments in their manufacture.

E.g. The spoon is orange in colour when cold but turns bright yellow if the food is too hot. Prior to using smart materials like these the consumer would have had to test the food/liquid by touching or tasting it. This is inaccurate and unhygienic.



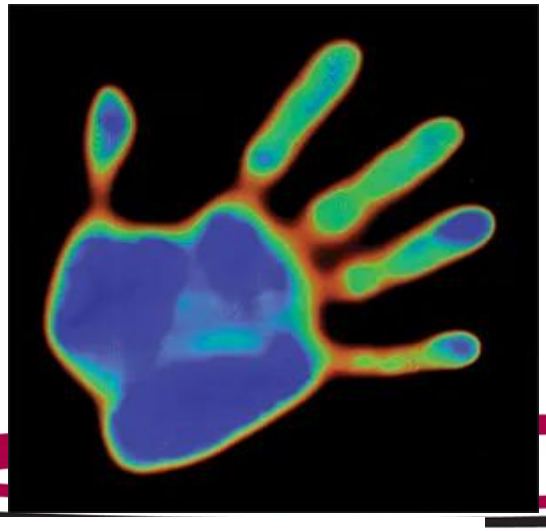


Video: Tommee Tippee Spoon



Applications - *Thermocolour Film*

Thermochromic pigments are a special type of liquid crystal which when heated will change colour. The introduction of these pigments in the form of an ink has led to the development of temperature testing strips for medical and other applications, such as testing safe water temperature in baths.



https://www.youtube.com/watch?v=DscgYJu_HGE



Temperature indicating bath mat.

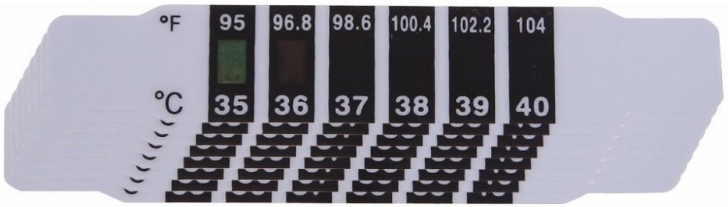
Video: Thermo-Chromic Sheet



Applications - *Thermometers*

Traditional mercury thermometers, whilst very accurate, are difficult to read and extremely dangerous to use with young children who may bite or break them.

Thermochromic pigments can be engineered to change colour across a range of temperatures.

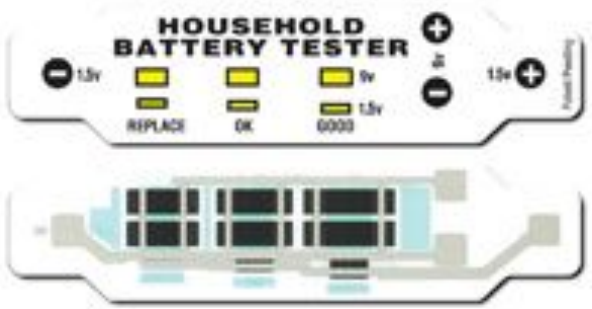




Applications - *Batteries*

Battery strips incorporate thermo-chromatic ink into a material that heats up as a current passes through. If the battery has sufficient energy to heat the strip, then it will change colour indicating the battery is in good condition.





Advantages of thermo-chromic materials

- The main characteristic of phosphorescent pigments is their capacity to absorb, store and emit light.
- After absorbing light they can glow in the dark for up to twelve hours.
- In addition the material is stable, non-toxic, has no radioactive additives and has good weatherability.

Disadvantages of thermo-chromic materials

- Certain chemicals such as photochromic or thermochromic materials may well be difficult to break down in nature.

Quantum Tunnelling Composites

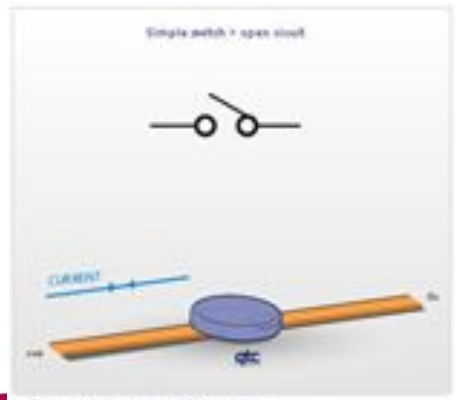
What are *'quantum tunnelling composites?'*

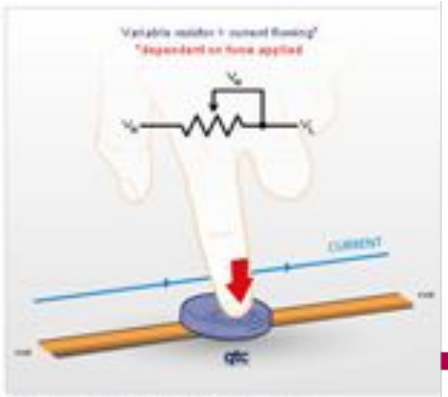
Quantum tunnelling composites (or QTCs) are composite materials of metals and non-conducting elastomeric binder (plastic and rubber), used as pressure sensors.

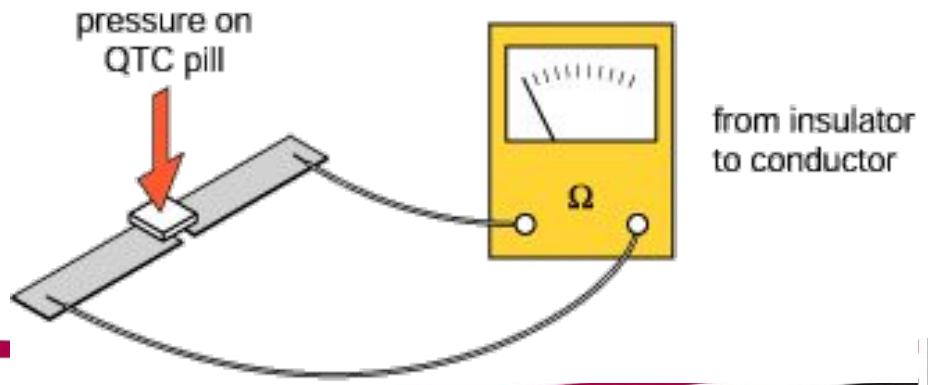
They utilise ***quantum tunnelling***: without pressure, the conductive elements are too far apart to conduct electricity; when pressure is applied, they move closer and electrons can tunnel through the insulator.

What are *'quantum tunnelling composites?'*

The effect is far more pronounced than would be expected from classical (non-quantum) effects alone, as classical electrical resistance is linear (proportional to distance), while quantum tunnelling is exponential with decreasing distance, allowing the resistance to change by a factor of up to 10^{12} between pressured and unpressured states.







App. - Quantum Tunnelling Composite Cord

This unassuming piece of cord is in fact a sample of Quantum tunnelling composite (QTC) - a flexible polymer that exhibits amazing and previously unseen electrical properties. In its standing state it is a perfect insulator, but when squeezed it dramatically changes to become a near-perfect conductor, through which very high currents are able to pass with very little resistance.



https://www.instituteofmaking.org.uk/materials-library/material/quantum-tunnelling-composite-cord

Advantages Quantum tunnelling composite

- QTC is a pressure/force sensing material. It can be easily integrated into existing products to enable force sensing opportunities and solutions.
- QTC material can be formed or moulded into virtually any size, thickness or shape, permitting redesign of product interfaces and providing improved ergonomics, aesthetics and user comfort.
- QTC material is durable – it has no moving parts to wear out.
- QTC material is mechanically strong.

Advantages Quantum tunnelling composite

- QTC material can be made to withstand extreme temperatures limits.
- QTC material is versatile, both electrically and physically e.g. Its range and sensitivity can be altered. QTC material is also intrinsically safe – the material is a contactless switch, ideal for sparkless operation.

Disadvantages Quantum tunnelling composite

- QTC acts a complete conductor after a certain level of compression reducing its effectiveness as a force sensor beyond that stage.

- QTC takes considerable time to return to its original state after deformation due to force.

Video: Quantum tunnelling composite



Quantum Tunnelling Composite Questions

1. Describe ***TWO*** features of a Quantum Tunnelling Composites (4)
2. State ***TWO*** advantages of Quantum Tunnelling Composites (2)
3. State ***TWO*** disadvantages of Quantum Tunnelling Composites (2)
4. Explain ***TWO*** Quantum Tunnelling Composite applications in micro-electronics (4)
5. Explain how Quantum Tunnelling Composite use impacts on the manufacture of products (2)